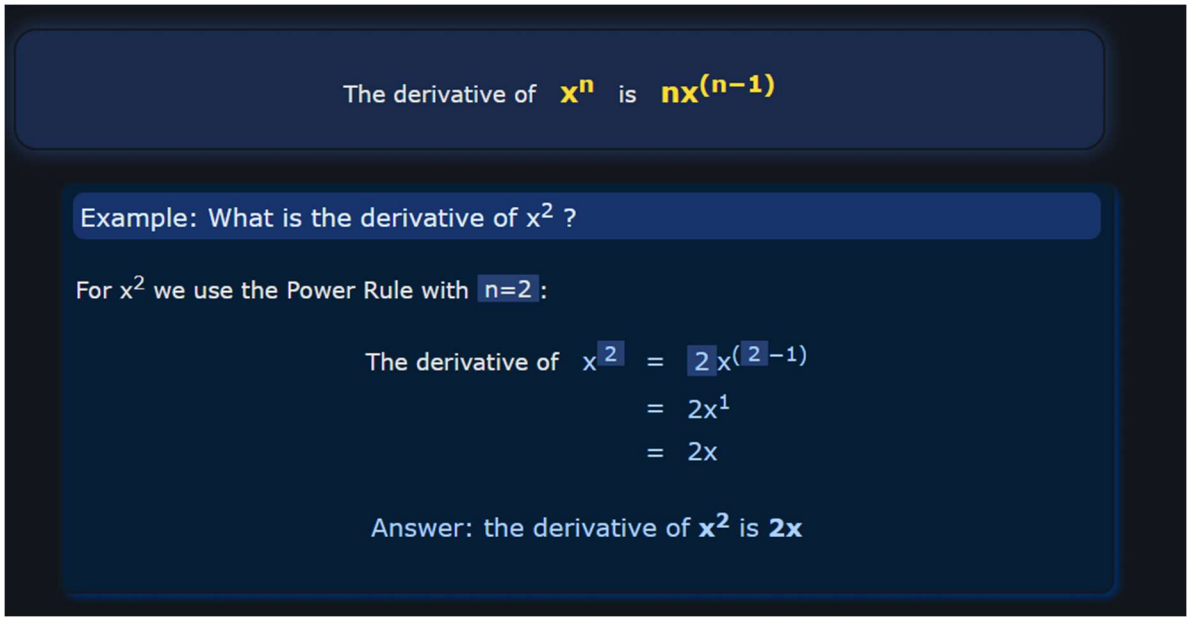
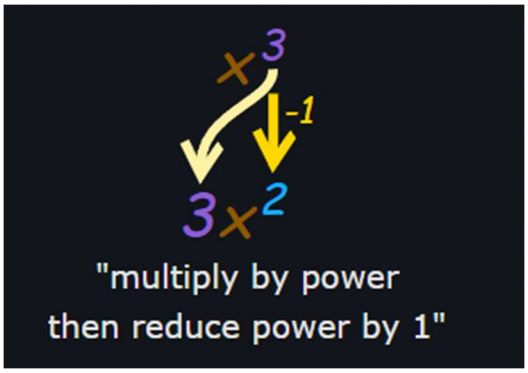
Derivative Rules

There are rules we can follow to find many derivatives

Rules	Function	Derivative
Multiplication by constant	cf	cf'
Power Rule	x^n	nx^{n-1}
Sum Rule	$f + g$	$f' + g'$
Difference Rule	$f - g$	$f' - g'$
Product Rule	fg	$f\,g' + f'\,g$
Quotient Rule	f/g	$\frac{f'\,g - g'\,f}{g^2}$
Reciprocal Rule	$1/f$	$-f'/f^2$

Power Rule





Multiplication by constant

Multiplication by constant

Example: What is $\frac{d}{dx}5x^3$?

the derivative of cf = cf′

the derivative of 5f = 5f′

We know (from the Power Rule):

$$\frac{d}{dx}x^3 = 3x^{3-1} = 3x^2$$

So:

$$\frac{d}{dx}5x^3 = 5\frac{d}{dx}x^3 = 5 \times 3x^2 = \mathbf{15x^2}$$

Sum Rule

Sum Rule

Example: What is the derivative of x^2+x^3 ?

The Sum Rule says:

$$\text{the derivative of } f + g = f' + g'$$

So we can work out each derivative separately and then add them.

Using the Power Rule:

- $\frac{d}{dx}x^2 = 2x$
- $\frac{d}{dx}x^3 = 3x^2$

And so:

$$\text{the derivative of } x^2 + x^3 = \mathbf{2x + 3x^2}$$

Difference Rule

Difference Rule

What we differentiate with respect to doesn't have to be **x**, it could be anything. In this case **v**:

Example: What is $\frac{d}{dv}(v^3 - v^4)$?

The Difference Rule says

$$\text{the derivative of } f - g = f' - g'$$

So we can work out each derivative separately and then subtract them.

Using the Power Rule:

- $\frac{d}{dv}v^3 = 3v^2$
- $\frac{d}{dv}v^4 = 4v^3$

And so:

$$\text{the derivative of } v^3 - v^4 = \mathbf{3v^2 - 4v^3}$$

Sum, Difference, Constant Multiplication And Power Rules

Example: What is $\frac{d}{dz}(5z^2 + z^3 - 7z^4)$?

Using the Power Rule:

- $\frac{d}{dz}z^2 = 2z$
- $\frac{d}{dz}z^3 = 3z^2$
- $\frac{d}{dz}z^4 = 4z^3$

And so:

$$\frac{d}{dz}(5z^2 + z^3 - 7z^4) = 5 \times 2z + 3z^2 - 7 \times 4z^3$$
$$= \mathbf{10z + 3z^2 - 28z^3}$$

Product Rule

Example: What is the derivative of cos(x)sin(x) ?

The Product Rule says:

$$\text{the derivative of } fg = f\,g' + f'\,g$$

Quotient Rule

To help you remember:

$$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

Reciprocal Rule

Example: What is $\frac{d}{dx}(1/x)$?

The Reciprocal Rule says:

$$\text{the derivative of } \frac{1}{f} = \frac{-f'}{f^2}$$

With f(x)= x, we know that f'(x) = 1

So:

$$\text{the derivative of } \frac{1}{x} = \frac{-1}{x^2}$$

Which is the same result we got above using the Power Rule.

Derivatives of some common Functions

Common Functions	Function	Derivative
Constant	c	0
Line	x	1
	ax	a
Square	x^2	$2x$
Square Root	$\sqrt{x}$	$(½)x^{-½}$
Exponential	e^x	e^x
	a^x	$\ln(a)\, a^x$
Logarithms	$\ln(x)$	$1/x$
	$\log_a(x)$	$1 / (x \ln(a))$
Trigonometry (x is in radians)	$\sin(x)$	$\cos(x)$
	$\cos(x)$	$-\sin(x)$
	$\tan(x)$	$\sec^2(x)$
Inverse Trigonometry	$\sin^{-1}(x)$	$1/\sqrt{(1-x^2)}$
	$\cos^{-1}(x)$	$-1/\sqrt{(1-x^2)}$
	$\tan^{-1}(x)$	$1/(1+x^2)$